

Compiler

--- Course Projects

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Projects

Project1: Lexical Analyzer Implementation

Option 1. X-LEX (100 points)

Option 2. XL-Scanner (50 points)

Project2: Syntax Parser Implementation

Option 1. Recursive-Decent based Parser (50 points)

Option 2. LR(1) based Parser (50 points)

Notices: 1)The programming language is not limited .

Requirements on X-LEX

Implement a software similar to LEX,

1. Input

(1) REs %%Token Patterns in a .l file

2)Output

(2) a program recognizing tokens defined by the REs.

3)Error handling may be included.

Requirements on XL-Scanner

1. Define REs by yourself for the common tokens used in our programming languages.
2. Convert REs into NFAs.
3. Merge these NFAs into a single NFA.
4. Convert the NFA into a DFA with minimum states.
5. Develop a scanner based on the DFA.

Requirements on Development of a Recursive- Decent based Parser

1. Input

- (1) Stream of characters
- (2) A CFG based on the tokens defined in the project 1.

2. Output

Sequence of actions in the recursive-decent parsing.

Requirements on Development of a LR(1) based Parser

1. Design a CFG that should describe the main sentence patterns in programming languages by using the tokens defined in the project 1.
2. Construct LR(1) parsing table for the CFG.
3. Implement the parser based on the LR(1) parsing table.

Document Requirements

1. Reports on labs (One report for each project.)

- a) Motivation/Aim
- b) Content description
- c) Ideas/Methods
- d) Assumptions
- e) Related FA descriptions
- f) Description of important Data Structures
- g) Description of core Algorithms
- h) Use cases on running
- i) Problems occurred and related solutions
- j) Your feelings and comments

2. CD on labs

- a) A CD is needed by a class.
- b) Each student owns a directory in the CD, and each project has a subdirectory.
- c) In each subdirectory, the following files are needed:
 - Input file, Output file, Source program files, Report file, Other related files

Submission Policies

1. **Deadline**: The day before the final exam.
2. Send to the teaching assistant via QQ.
3. Convert your directory into a compressed file named using **student ID+name**.
4. You can submit your compressed file in advance.